

EXPERIMENT: GYM FITNESS TRACKER VISUALIZATION

AIM:

To create data visualizations such as pie charts and bar graphs to analyze workout activities and calories burned using JavaScript and Chart.js.

OBJECTIVE:

- To understand data visualization techniques
- To represent fitness data using charts
- To use Chart.js for creating interactive graphs
- To display values inside a pie chart

INTRODUCTION:

Data Visualization is the graphical representation of data using charts and graphs. In this experiment, a Gym Fitness Tracker is developed to visualize workout data such as calories burned across different activities like Cardio, Strength Training, Yoga, HIIT, and Cycling.

PROCEDURE:

Step 1: Create HTML File (index.html)

- Create a basic HTML structure
- Add canvas elements for charts
- Include Chart.js and datalabels plugin

Step 2: Create JavaScript File (script.js)

- Define workout data
- Configure pie chart with values inside
- Configure bar chart

Step 3: Run the Project

- Open index.html in a browser

CODE:

Refer to index.html and script.js files created using Chart.js and datalabels plugin.

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width,
initial-scale=1.0">

  <title>Gym Fitness Tracker</title>

  <style>

    body {

      font-family: Arial;

      text-align: center;

      margin: 40px;

      background-color: #f4f6f9;

    }

    h1 {

      color: #333;

    }

    canvas {

      margin: 20px auto;

      display: block;

      background: white;

      padding: 15px;
```

```
        border-radius: 12px;

        box-shadow: 0 0 10px rgba(0,0,0,0.1);

    }

</style>
</head>
<body>

    <h1>Gym Fitness Tracker</h1>

    <!-- Pie Chart -->

    <canvas id="pieChart" width="400" height="400"></canvas>

    <!-- Bar Chart -->

    <canvas id="barChart" width="400" height="400"></canvas>

    <!-- Chart.js -->

    <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>

    <!-- Plugin for showing values -->

    <script
src="https://cdn.jsdelivr.net/npm/chartjs-plugin-datalabels"></scrip
t>

    <!-- JS File -->

    <script src="script.js"></script>
```

```
</body>
```

```
</html>
```

```
// Register plugin
```

```
Chart.register(ChartDataLabels);
```

```
// Fitness Data (Calories Burned)
```

```
const fitnessData = {
```

```
  labels: ['Cardio', 'Strength', 'Yoga', 'HIIT', 'Cycling'],
```

```
  datasets: [{
```

```
    label: 'Calories Burned',
```

```
    data: [300, 450, 200, 350, 250],
```

```
    backgroundColor: [
```

```
      '#ff6384',
```

```
      '#36a2eb',
```

```
      '#ffce56',
```

```
      '#4bc0c0',
```

```
      '#9966ff'
```

```
    ]
```

```
  }]
```

```
};
```

```
// PIE CHART (with values inside)
```

```
const pieCtx = document.getElementById('pieChart').getContext('2d');
```

```
new Chart(pieCtx, {

  type: 'pie',

  data: fitnessData,

  options: {

    plugins: {

      title: {

        display: true,

        text: 'Workout Distribution (Calories)'

      },

      datalabels: {

        color: '#fff',

        font: {

          weight: 'bold',

          size: 14

        },

        formatter: (value) => value + ' cal'

      }

    }

  }

});

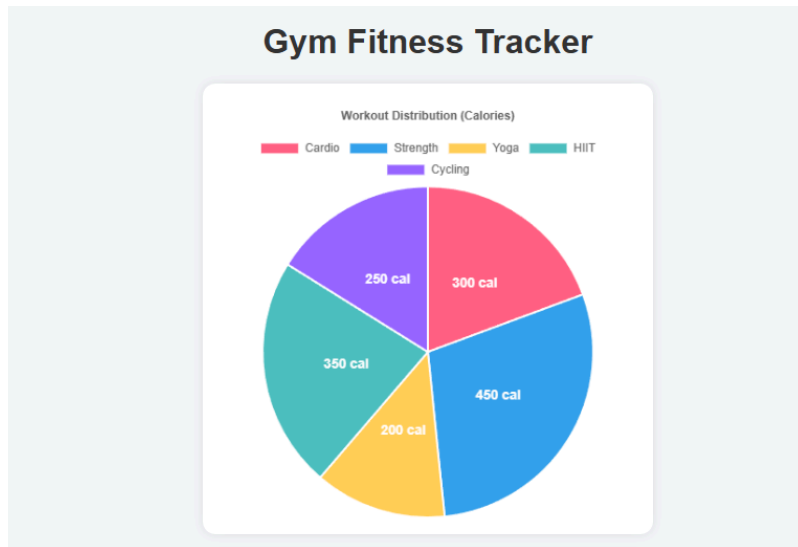
// BAR CHART

const barCtx = document.getElementById('barChart').getContext('2d');
```

```
new Chart(barCtx, {  
  
  type: 'bar',  
  
  data: fitnessData,  
  
  options: {  
  
    plugins: {  
  
      title: {  
  
        display: true,  
  
        text: 'Calories Burned by Workout'  
  
      }  
  
    },  
  
    scales: {  
  
      y: {  
  
        beginAtZero: true  
  
      }  
  
    }  
  
  }  
  
});
```

OUTPUT:

- Pie chart showing calorie distribution (values inside chart)



- Bar chart comparing workout activities



RESULT:

Thus, the Gym Fitness Tracker visualization was successfully created using JavaScript and Chart.js with values displayed inside the pie chart.